

AWS ACADEMY PROJECT PROOF

Microservices and CI/CD Pipeline Builder

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This document contains screenshots collected from the completed AWS Academy lab environment as evidence of the implemented microservices, container registry, ECS deployment, CI/CD pipeline, blue/green deployment, database integration, and final service status.

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Evidence 1: Course Completion Certificate



AWS Academy Graduate – Microservices and CI/CD Pipeline Builder certificate.

Evidence 2: AWS Cloud9 Development Environment

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 8508-7779-2031 voclabs/user4854361=Megha

AWS Cloud9 > Environments

My environments
Shared with me
All account environments
Documentation

Environments (1) [Delete] [View details] [Open in Cloud9] **Create environment**

My environments

	Name	Cloud9 IDE	Environment type	Connection	Permission	Owner ARN
☐	MicroservicesIDE	Open	EC2 instance	Secure Shell (SSH)	Owner	arn:aws:sts::850877792031:assumed-role/voclabs/user4854361=Megha

AWS Cloud9 environment used for developing and managing the microservices.

Evidence 3: Supplier Database – Amazon RDS / MySQL

```
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.03 sec)

mysql> USE COFFEE;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_COFFEE |
+-----+
| suppliers |
+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM suppliers;
+----+-----+-----+-----+-----+-----+
| id | name   | address   | city       | state   | email |
+----+-----+-----+-----+-----+-----+
| 1  | Megha  | Btm       | Bangalore  | Karnataka | megha.saha2409@g |
| 2  | Sabeena | Whitefield | Los Angeles | US       | nwolf@example.co |
+----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> exit
Bye
ubuntu@ip-10-16-10-11:~/resources/codebase partner$
```

Supplier database output showing the suppliers table and records used by the Employee Microservice.

Evidence 4: Amazon ECR Repositories

The screenshot displays the Amazon Elastic Container Registry (ECR) console. The left sidebar shows the navigation menu with options for Private registry, Public registry, and ECR public gallery. The main content area shows the Private repositories page with a table listing two repositories: 'customer' and 'employee'. The table columns are Repository name, URI, Created at, Tag immutability, and Encryption type. The 'customer' repository was created on September 09, 2026, at 10:27:44 (UTC+05.5) and is mutable with AES-256 encryption. The 'employee' repository was created on September 09, 2026, at 10:28:07 (UTC+05.5) and is also mutable with AES-256 encryption.

Repository name	URI	Created at	Tag immutability	Encryption type
customer	850877792031.dkr.ecr.us-east-1.amazonaws.com/customer	September 09, 2026, 10:27:44 (UTC+05.5)	Mutable	AES-256
employee	850877792031.dkr.ecr.us-east-1.amazonaws.com/employee	September 09, 2026, 10:28:07 (UTC+05.5)	Mutable	AES-256

Separate Amazon ECR repositories for the Customer and Employee container images.

Evidence 5: Amazon ECS – Final Service Status

The screenshot displays the Amazon Elastic Container Service (ECS) console for a cluster named 'microservices-serverlesscluster'. The cluster is in an 'Active' state, with 2 active services and 4 running tasks. The 'Services' tab is selected, showing a list of two services: 'customer-microservice' and 'employee-microservice'. Both services are in an 'Active' state and have 3/3 tasks running. The console also shows the cluster overview, including the cluster ARN, status, CloudWatch monitoring, and action logs.

Cluster overview

- Cluster ARN: `arn:aws:ecs:us-east-1:85087779:...`
- Status: Active
- CloudWatch monitoring: Default
- Action logs: Turned off
- Services: 2 active | 0 draining
- Tasks: 4 running | 0 pending
- Registered container instances: -

Services (2)

Service name	ARN	Status	Scheduling strategy	Task definition	Deployments and tasks
customer-microservice	<code>arn:aws:ecs:us-east-1:85087779:service/customer-microservice</code>	Active	Replica	customer-microservice	3/3 Tasks running
employee-microservice	<code>arn:aws:ecs:us-east-1:85087779:service/employee-microservice</code>	Active	Replica	employee-microservice	1/1 Tasks running

Final ECS cluster showing customer-microservice and employee-microservice running successfully.

Evidence 6: Customer CI/CD Pipeline

The screenshot displays the AWS CodePipeline console interface. At the top, the navigation bar includes the AWS logo, a search bar, and account information (United States (N. Virginia), Account ID: 8508-7779-2031, user: voclabs/user4854361=Megha). The breadcrumb trail shows the path: Developer Tools > CodePipeline > Pipelines > update-customer-microservice. The main header for the pipeline 'update-customer-microservice' features tabs for Pipeline, Executions, Triggers, Settings, and Tags, along with action buttons: Edit, Stop execution, Create trigger, Clone pipeline, and Release change. The 'Pipeline' tab is active, showing a visual representation of the pipeline with two stages: 'Source' and 'Deploy'. Both stages are marked as successful with green checkmarks. The 'Source' stage includes actions for 'AWS CodeCommit' and 'Amazon ECR'. The 'Deploy' stage includes an action for 'Amazon ECS (Blue/Green)'. The pipeline is currently in a 'Horizontal' view.

update-customer-microservice CodePipeline showing successful Source and Deploy stages.

Evidence 7: AWS CodeDeploy – Blue/Green Deployment

The screenshot displays the AWS CodeDeploy console for the application 'microservices-app'. The left sidebar shows the navigation menu with 'Developer Tools' expanded, leading to 'CodeDeploy'. The main content area shows the application details, including the name 'microservices-app' and the compute platform 'Amazon ECS'. Below this, the 'Deployment groups' tab is selected, showing a table of deployment groups. The table lists two groups: 'microservices-customer' and 'microservices-employee', both with a status of 'Succeeded'. The 'Last attempted deployment' and 'Last successful deployment' columns show the same timestamp for both groups, indicating a successful deployment. The 'Trigger count' column shows 0 for both groups.

microservices-app [Notify](#) [Delete application](#)

Application details

Name: microservices-app Compute platform: Amazon ECS

Deployments **Deployment groups** Revisions

Deployment groups [View details](#) [Edit](#) [Create deployment group](#)

Q

	Name	Status	Last attempted deploym...	Last successful deployment	Trigger count
☐	microservices-customer	✔ Succeeded	Sep 9, 2026 8:00 PM (UTC...	Sep 9, 2026 8:00 PM (UTC...	0
☐	microservices-employee	✔ Succeeded	Sep 9, 2026 9:01 PM (UTC...	Sep 9, 2026 9:01 PM (UTC...	0

microservices-app deployment groups showing successful Customer and Employee deployments.

Evidence 8: Employee Microservice Output

← → ↻ ⚠ Not secure 13.220.240.229:8081/admin/suppliers ☆ M ⋮



Manage coffee suppliers

Administrator home
Customer home

All suppliers

Name	Address	City	State	Email	Phone	
Megha	Btm	Bangalore	Karnataka	megha.saha2409@gmail.com	4155551212	edit
Sabeena	Whitefield	Los Angeles	US	nwolf@example.com	4155551214	edit

[Add a new supplier](#)

Employee supplier-management interface at the /admin/suppliers route.

Evidence 9: Final ECS Verification

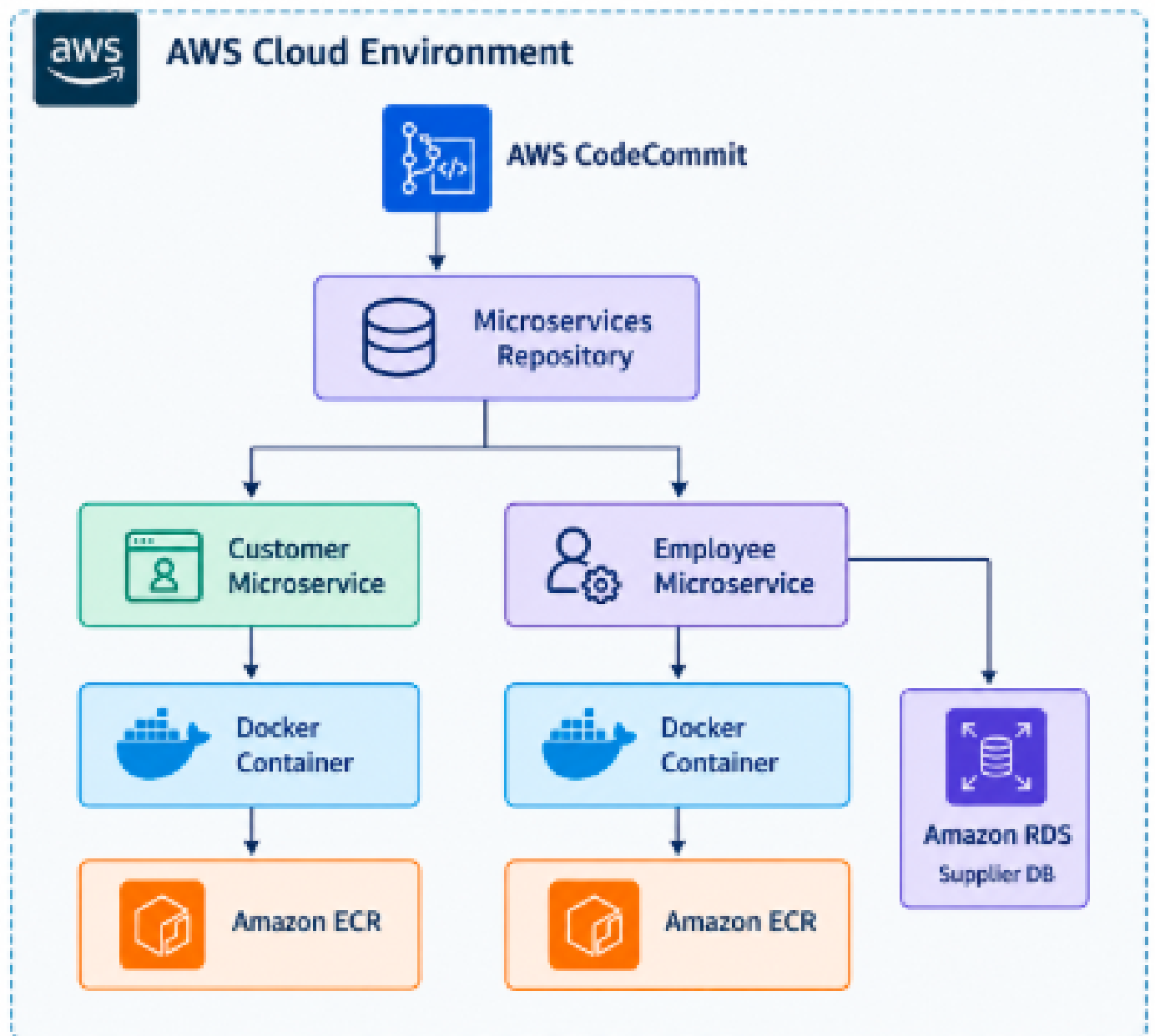
```
voclabs:~/environment $ aws ecs describe-services \
  --cluster microservices-serverlesscluster \
  --services customer-microservice employee-microservice \
  --query 'services[*].[serviceName,desiredCount,runningCount,status]' \
  --output table \
  --no-cli-pager
```

DescribeServices			
customer-microservice	3	3	ACTIVE
employee-microservice	1	1	ACTIVE

```
voclabs:~/environment $
```

Terminal verification showing both ECS services ACTIVE with the final running task counts.

Evidence 10: Project Architecture



Architecture used for the AWS Academy microservices implementation.

Proof Summary

- Customer and Employee microservices were containerized and stored in separate Amazon ECR repositories.
- The services were deployed on Amazon ECS using AWS Fargate.
- AWS CodePipeline and AWS CodeDeploy were used for CI/CD and blue/green deployment.
- The Employee Microservice accessed supplier records through the database configuration.
- Final ECS verification showed 3 running Customer tasks and 1 running Employee task.